

Differential Privacy and an Analysis of Feature-Label Correlations for Label Inference Protection

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Abstract

To better understand privacy mechanisms, this research explores label inference attacks within the framework of differential privacy, specifically examining the vulnerability of datasets with privatized labels when its feature set remains public. While differential privacy mechanisms like label aggregation and Laplacian/Gaussian noise additions provide adversarial resistance against certain attacks, namely Membership Inference Attacks and Model Inversion attacks, an adversary with access to public features can still recover private labels with high accuracy through Correlation Inference Attacks and Simple Prediction Attacks. This study evaluates an extension of label inference attacks on the Adult Census Income dataset using Random Forest classifiers trained on all possible feature subsets (2^{10} models), measuring discriminative advantage between each classifier subset through AUC score analysis. Findings have shown that complete feature release could result in high probability of label disclosure when feature-label correlations are strong. Therefore, determining which public features require differential privacy protection is critical for effective risk assessment and for maintaining an appropriate balance between privacy and utility.

1 Introduction

1.1 Problem Statement

Dataset sharing for machine learning creates a fundamental privacy challenge. Specifically, how can organizations release sensitive data for model training while protecting the Personally Identifiable Information (PII) contained within that data? This question becomes increasingly complex when we consider that not all information in a dataset is equally sensitive. In dataset contexts regarding medical, credit, or income related information, its label set (e.g., whether a patient has a disease, whether a loan should be approved, or whether someone makes over \$50K annually) are data elements that are considered highly sensitive PII. In contrast, the features (e.g., age, education level, employment status) may be considered less sensitive and could be more

ignorantly released without the need of a privacy mechanism.

This is exemplified by a relaxation of differential privacy, Label-DP, which describes a privacy mechanism that works in protecting only the sensitive labels while allowing features to remain public. The intuition behind Label-DP mechanisms like Randomized Response (RR) is that the feature set can be released publicly if we randomly flip values in its label set [1]. RR enables plausible deniability since labels are noisy at the individual level, but statistical relationships between aggregates of features and labels remain, enabling models like Random Forest to still learn underlying relationships. Therefore RR enables utility of datasets to be preserved, however, researchers have discovered that RR privacy mechanisms are vulnerable to threat models that work with a publicized feature matrix [2]. This is because useful models will inherently reveal labels given the features it was trained with, as the model understands associations that should be made between the feature space and the target variable. As a result, this can reverse noise added privacy schemes.

1.2 Objectives

This project focuses on identifying feature-label correlations of an Adult Census Income Dataset. Ten features, encompassing: age, workclass, education, marital status, occupation, relationship, race, sex, hours worked per week, and native country, will be used to predict whether or not an individual has a salary above \$50,000 a year. The study will quantify feature-label correlations by training a classifier on every possible subset of the ten available features, training a total of 1024 models. By evaluating the Area Under the Curve (AUC) metric for each subset, we can isolate which specific combinations of public attributes possess the highest predictive power in regards to the sensitive income label. A high AUC score for a given feature subset indicates a strong correlation, revealing that those specific features have strong determinant influence in reconstructing the private label with high confidence.

The research is aimed to answer how vulnerable are datasets with privatized labels to label inference attacks, and

how an adversary's choice of feature subsets affect its label reconstruction advantage. Therefore research will be able to achieve two things:

- Understand the correlation between feature-label relationships
- Identify which feature combinations enable the most effective label recovery

Overall, being able to aggregate data elements that are strong determinants of a privatized label will immensely aid in the acknowledgement of risk assessments for dataset releases.

2 Motivation

The main motivation of this research is to study label inference attacks involving feature subsets on a real-world dataset containing practical PII data elements. While prior work has established theoretical bounds on the label inference attack advantage under the typical differential privacy definition, less attention has been paid to understanding how an attacker might leverage feature selection strategies to enhance their attacks. Insight behind the rationale of how feature selection strategies might broaden an attack surface will be more formally discussed in the related work of Section 3.

Additionally, the Adult Census Income dataset will be utilized because:

- It contains categorical and continuous features
- The binary classification task of income being greater than or less than \$50K represents a realistic sensitive label
- The features have intrinsic and varying correlations with income (e.g. it is more statistically probable that an individual with higher education would have a higher salary), allowing us to study the most important feature correlations that would aid in identifying the level of an income an individual has

3 Related Work

This research builds upon the methodology of examining privacy mechanisms for machine learning, particularly in settings where labels are sensitive but features are public. Recent work has increasingly questioned whether traditional privacy metrics adequately capture the true privacy risks, leading to the development of more nuanced evaluation frameworks.

3.1 Label Differential Privacy

The concept of Label Differential Privacy is that it is a relaxation of standard differential privacy that protects only training labels while allowing features to be treated as public. This relaxation was motivated by applications where only the label is considered sensitive. The relevance of this relaxation is best exemplified by a model trained on advertisement features, with a privatized label modeling user click-through rate [2]. In terms of utility, it is understandable that there would be little drawback in releasing a dataset containing features that would be already publicly available (e.g. the type of advertisement showcased by a product). Fundamentally, Label-DP will be able to achieve better privacy-utility tradeoffs than full data DP, but use cases of Label-DP are highly dependent on how strong feature-label correlations are, as discussed in the following section.

3.2 Label Inference Attacks

Prior research by Busa-Fekete et al. (2023) has exposed critical vulnerabilities in randomized response (RR) as a label differential privacy mechanism [3]. Researchers have demonstrated that when features remain public, RR provides inadequate protection against label inference attacks, even with strong privacy parameters (low epsilon values). Their theorem stated a mechanism with a strong privacy budget can still return a high attack utility in cases where the dataset's features are a strong predictor of its label [3]. As such, noise adding schemes were not impactful in preventing label inference attacks. In response, they proposed Double Randomized Response (DRR), which randomizes both labels and features by replacing some data elements in the train set with synthetically generated data. The overarching idea is that DRR preserves utility without sacrificing privacy by ensuring the synthetic feature and label distributions remain statistically close to the true conditional distribution. Given the same adversarial model and using RR as baseline, DRR as a defense mechanism was shown to decrease label inference attack accuracy by almost 40%. And in terms of utility, DRR was shown to be only 5% less accurate than RR, showcasing that the DRR model is still able to generalize the underlying relationship between feature conditions exceedingly well.

However, DRR's reliance on synthetic data quality suggests a potential vulnerability: if feature-label correlations differ significantly between real and synthetic data, its Rényi divergence will be high, translating to a higher epsilon value, which in turn implies a weaker privacy budget. The basis for this research was theorized on this idea that by performing feature subset selection, we can

identify feature combinations that exhibit strong correlations with labels in the original data but are maybe poorly replicated in the synthetic distribution. Such subsets would demonstrate high discriminative power for label inference despite DRR's defenses. Essentially, if DRR fails to preserve these dependent relationships in synthetic data, an attacker can exploit this inconsistency (e.g. lack of feature-feature correlations) to distinguish real from synthetic examples and recover label information. By analyzing which feature subsets preserve strong feature-label correlations in the original data but not in the synthetic data, we can identify specific vulnerabilities in DRR's protection mechanism. However due to time constraints, this research has focused more on identifying feature-label correlations, on the basis of understanding what dataset features strongly predict its label.

3.3 Differential Privacy Paradox

Prior work by Wu et al. (2022) has also discussed different interpretations of what type of leakage Label-DP entails [2]. It is argued that attack utility should not be measured by the absolute accuracy of label inference attacks, rather the advantage of the attack utility in comparison to Bayes Classifier. Bayes Classifier was argued to be used as a baseline metric because it is based on only finding statistical relationships of dataset features in relation to a label.

By this baseline metric, two separate sources of label inference power are able to be separated:

- Statistical inference derived by correlation
- Extra leakage derived by memorizing the actual training labels

The overarching idea for my own research was partially inspired by the separation of powers of the formal definition. The latter bullet point describes scenarios where the dataset structure itself results in an attack advantage, but this research planned to explore the former bullet point by finding a privacy mechanism that could lower baseline leakage derived by statistical correlations between features and labels.

3.4 Correlation Inference Attacks

Prior work by Cretu et al. (2024) showcased developments of Correlation Inference Attacks (CIA), proposing two complementary CIA models that reveal correlation leakage. One of the models in particular focuses on exploiting the mathematical structure of correlation matrices [4]. This attack uses knowledge of correlation constraints (e.g. how strong a feature correlates with a label) to infer unknown

correlations between feature pairs. Correlation constraints are typically public knowledge, further contributing to the inspiration in my own analysis of feature-label correlation and their respective contribution in successfully classifying a label.

4 System Design

4.1 Dataset

The public Adult Census Income dataset from Kaggle will be utilized. The Dataset contains roughly 32,000 training samples and involves the binary classification task of predicting whether annual income exceeds \$50K or not.

The following ten features from the dataset were selected:

1. Age: Continuous variable representing age in years. Ages ranged from 17-90. A large proportion of individuals were aged 40 and up. Increasing age was initially predicted to be a strong predictor of annual income exceeding \$50K.
2. Workclass: Categorical variable representing employment sector (Private, Government, Self-employed, or other). Private and government employment sectors initially predicted to be a strong predictor of annual income exceeding \$50K.
3. Education: Categorical variable representing highest level of education (HS-grad, Some College, Bachelor's, Associates, Master's). Initially predicted higher levels of education to be strong predictors of annual income exceeding \$50K.
4. Marital-Status: Categorical variable representing marital status (Married, Nevermarried, Divorced, etc.). Initially predicted a married status to be a strong predictor of annual income exceeding \$50K.
5. Occupation: Categorical variable representing job type (Exec-managerial, Prof-specialty, Craft-repair, Sales, etc.).
6. Relationship: Categorical variable representing family relationship (Husband, Wife, Own-child, unmarried, etc.). Initially predicted own-children and unmarried relationship statuses to be strong predictors of annual income NOT exceeding \$50K.
7. Race: Categorical variable representing racial/ethnic category. Included because it is important to acknowledge racial wealth gaps.
8. Sex: Binary categorical variable (Male/Female). Included because it is important to acknowledge gender wage gaps.

9. Hours-per-Week: Continuous variable representing average hours worked per week. Initially predicted a higher value to be a strong predictor of annual income exceeding \$50K.
10. Native-Country: Categorical variable representing country of origin (USA, Mexico, Philippines, Germany, etc.).

4.2 System Diagram

The research employs a two phase design to evaluate feature-label correlations in datasets with privatized labels.

The Data Set Preparation phase begins with the Adult Census Income dataset and drops certain features, particularly:

- Fnlwgt: describes the portion of individuals in the actual U.S. population the row is representative of, removed because it was predicted to distort socioeconomic correlations between feature sets. Preferred each row entry to have equal weight of consideration.
- Education number: describes number of years of education an individual has obtained, removed because it is a duplicate of education
- Capital gain/loss: describes profit/loss from selling an asset. Capital gain had a value of 0 for most entries, so as a result, both metrics were removed from the dataset

Categorical variables are one-hot encoded, and the resulting feature space is categorized into a consolidated set of 10 primary feature vectors. The dataset is then split into 70% training and 30% test sets, with graphical distributions displayed to characterize the feature space before model training.

The Model Training (MT) phase implements an exhaustive subset selection strategy: for each subset size k ranging from 1 to 11, the system iterates through all unique feature combinations of that size, training an independent Random Forest classifier on each subset. For every trained model, the system extracts and records the AUC score on the test set, accumulating results for over 1,024 total models. Table 1 below showcases the number of models created across each k -feature, a behavior modeled by their binomial coefficients.

The aim of this exhaustive approach was to enable the identification of specific feature combinations that exhibit the highest predictive power for label inference. By analyzing AUC distributions across all feature subsets, the research can quantify the discriminative advantage between optimal feature combinations, revealing gaps in privacy

mechanisms. Figures 1 and 2 below summarizes the two phases of the system diagram.

Feature Subset Size	Number of Models
1	10
2	45
3	120
4	210
5	252
6	210
7	120
8	45
9	10
10	1

Table 1. Number of Models Per K-Feature-Subset

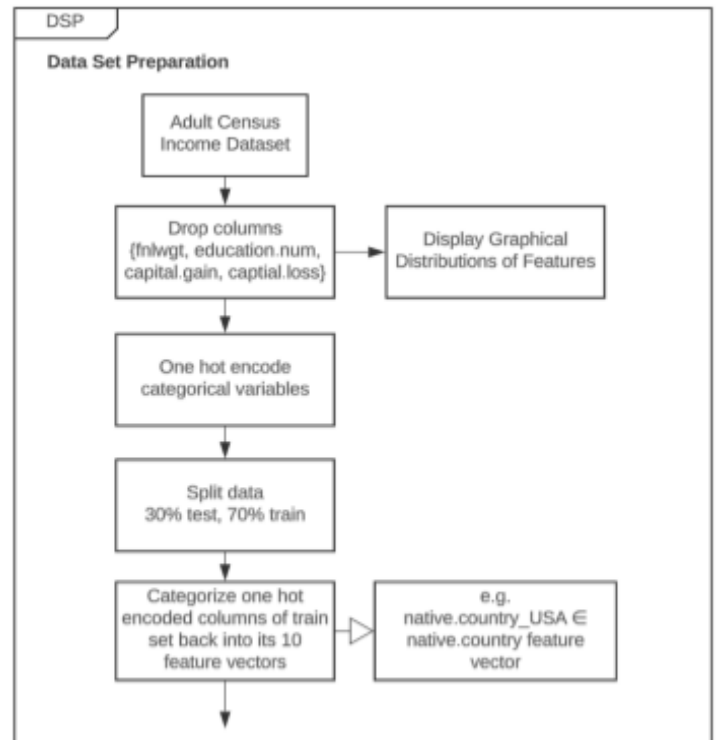


Figure 1. Data Set Preparation

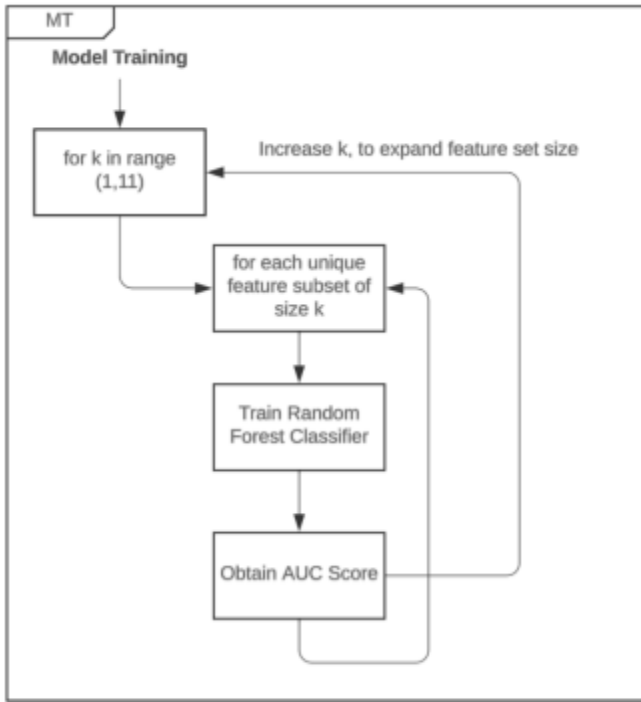


Figure 2. Model Training

5 Experimental Results

After obtaining all AUC scores from the 1,024 trained Random Forest models, a comprehensive set of visualizations were created to better understand different trends about feature-label correlations:

- 1) A scatter plot that organizes all feature combinations by size to show how prediction accuracy tend to improve as more features are added
- 2) A trendline graph that traces how the best, average, and worst performing feature combinations change as more features are added. Slope differences between points on trend line help enunciate time (enumeration) stamps where adding more features becomes impactful
- 3) A frequency bar chart that identifies which specific features appear most often in the highest-performing combinations
- 4) A summary table that lists the combination of features at each k value that produced the max AUC score

5.1 AUC Scores for All Feature Combinations

Appendix A.1 showcases the analysis of all 1,024 feature subset combinations via a scatterplot. When plotting their

position in ascending order of k, the scatter plot demonstrates a stratification of AUC scores by feature subset size. The plot helps in highlighting the stratification through different dot colors per k value. Single feature subsets are clustered tightly in the 0.50–0.75 AUC range, indicating that individual features alone provide only moderate predictive power of labels. As subset size increases (k=8,9,10), the scatter progressively shifts upward, concentrating near a peak of 0.90 AUC.

5.2 AUC Score Trendline

Appendix A.2 showcases the trendline analysis across the k-feature subsets, demonstrating nonlinear growth in maximum, mean, and minimum AUC performance as feature subset size increases. The red line that describes maximal AUC performance shows the steepest initial growth, rising from 0.7876 at k=1 to 0.8956 at k=9. More importantly, max AUC score slightly declines to 0.8954 at k=10.

Since the ninth-best feature combination achieves marginally better performance than the feature combination encompassing the entire set of features, there could be intrinsic correlations between certain feature subsets that are more determinant of label inferences. This idea will be explored in more detail in Section 5.5 when analyzing the table of features with best determinant strength.

The blue line that describes mean AUC performance grows more logarithmically, reflecting that increases in feature set provides diminishing returns. Specifically, adding features beyond k=6 provides minimal improvement (less than 0.03 AUC gain in the average case). The green line that describes minimum AUC performance showcases that worst case feature subsets of larger cardinality still outperform the best single feature models. The shaded region between minimum and maximum AUC expands dramatically from k=1 to k=4, but then tightens from k=5 onward, showing that larger feature sets exhibit more consistent behavior in predictive power.

5.3 Most Frequent Features in Highest AUC Score

Appendix A.3 showcases a frequency analysis among the top 50 performing subsets to reveal disparities in which features contribute the most to the best Random Forest classifiers. Three features, as shown in red, appear with 100% frequency in the top 50 subsets: education, occupation, and age, indicating these attributes are essentially mandatory for achieving maximal predictive power. Hours per week, relationship, and marital.status appear in 96%, 70%, and 68% of the top 50 subsets, respectively. Workclass, sex, and native country appear in

54% of top subsets, indicating moderate contribution. Finally, race appears in only 48% of the top 50 subsets, making it the least essential demographic attribute for income prediction amongst this dataset's features. This frequency metric provides insight that: in relation to this dataset, privacy mechanisms should prioritize the protection of education, occupation, and age due to their strong predictive power, as indicated by their 100% contribution in the top 50 feature subsets. The finding also suggests that an adversary performing targeted feature selection would focus on combinations heavily weighted toward education, occupation, and age, if creating a minimal feature set with the most efficient discriminative power.

5.4 Features with Best Determinant Strength

k	Max AUC	Mean AUC	Features of Max AUC
1	0.78764	0.66576	{relationship}
2	0.85960	0.75086	{education, relationship}
3	0.87314	0.79980	{education, occupation, relationship}
4	0.88077	0.83061	{hour per week, education, occupation, relationship}
5	0.88891	0.85174	{age, hours per week, education, occupation, relationship}
6	0.89250	0.86715	{age, hours per week, workclass, education, occupation, relationship}
7	0.89433	0.87800	{age, hours per week, workclass, education, occupation, relationship, sex}
8	0.89498	0.88595	{age, hours per week, workclass, education, marital status, occupation, race, sex}
9	0.89568	0.89150	{age, hours per week, workclass, education, marital status, occupation, race, sex, native country}
10	0.89542	0.89542	{age, hours per week, workclass, education, marital status, occupation, relationship, race, sex, native country}

Table 2. Best AUC Score for K-Features

Table 2 above showcases an enumeration of the optimal feature subsets across all k values. The results demonstrate that in the best case, label disclosure can be attributed to a small subset of features. Specifically, education, occupation, age, hours per week, and relationship were main factors in driving inference capability. Moreover, the k=5 optimal subset contained these five features, achieving .88891 AUC, and effectively recovering 99.3% of the overall best classifier's discriminative power (0.89542 AUC at k=10) while excluding five other features entirely (marital status, race, sex, workclass, native country).

As a side note, it is somewhat surprising that the relationship feature was the only feature that appeared in all subsets k=1 to k=5, considering it was only in 70% of the top 50 AUC scores as depicted in Appendix A.3 This observation requires a more in depth analysis to be explained, as it would be expected that one of {age, occupation, education} would be the main driving factor for feature-label relationship due to them being features in 100% of the top 50 feature subsets.

The table's finding has implications for differential privacy mechanisms. Particularly, the conventional Label-DP definition implies that releasing all features alongside privatized labels is acceptable in achieving differential privacy, yet this analysis shows that an adversary exploiting feature selection mechanisms can achieve near-maximal label inference using only half the available feature set. In the best case, the marginal AUC gains diminish sharply beyond k=5, indicating that additional features may not enable models to gain a better understanding of the feature-label relationship.

6 Limitations

This research, while providing valuable insights into feature based label inference vulnerabilities, operates within several important constraints.

- Analysis focuses exclusively on the Adult Census Income dataset, which exhibits specific statistical properties (moderate class balance, categorical and continuous features, well defined feature-label correlations) that may not generalize to other datasets. Particularly, datasets with extreme value distributions may exhibit different KthSubset-AUC relationships.
- Research measures label inference capability through AUC scores on test sets, representing the threat model where an adversary trains a model and evaluates on known data.

- Threat model assumes the adversary has full knowledge of which features are available and has the ability to freely construct feature subsets arbitrarily.
- Research does not evaluate feature subset selection attacks against Double Randomized Response (DRR) or other more advanced privacy mechanisms that use synthetic data generation. While Section 3.2 motivates the hypothesis that DRR may be vulnerable to subset selection attacks due to imperfect synthetic data quality, the validation of this hypothesis remains for future work. The current study establishes a baseline understanding of which feature combinations are most dangerous for label inference on non-privatized data. It remains to be seen whether the feature distribution synthetically created by DRR can successfully prevent attacks involving advanced correlation analysis.
- Threat model assumes the adversary's knowledge of the underlying label distribution is accurate. In reality, estimating this distribution from an auxiliary dataset will degrade attack performance.

7 Conclusions

This research showcases that through exhaustive enumeration of all 1,024 possible feature combinations on the Adult Census Income dataset, adversaries may not need to use all publicly released features to achieve high label inference accuracy. Specifically, a five feature subset comprising education, occupation, age, hours per week, and relationship had achieved 99.3% of what the maximal AUC score had achieved utilizing the entire feature set. This finding challenges the conventional Label-DP definition that guarantees PII privacy protection being through privatized labels.

Furthermore, graphical frequency analysis revealed that education and occupation appear in 100% of top 50 performing subsets while race appears in only 48%, implying that privacy risk is not homogeneous across the feature space. This finding has immediate practical implications as not all features pose equal threat to label privacy. Privacy mechanisms should move toward more nuanced approaches that account for feature-label correlation strength and its combinatorial implications. Our analysis suggests three risk assessment related policy mechanisms:

- 1) Apply differential privacy selectively to high value features rather than uniformly to a feature set

- 2) Establish controls that restrict which feature combinations can be accessed together, preventing adversaries from constructing specific high AUC subsets identified in Section 5.5
- 3) Perform feature-to-feature privacy auditing, similar to the methodology described in this research, as to identify feature combinations that pose the greatest label reconstruction risk

Strategic protection of the core feature set could significantly reduce an adversary's attack surface without sacrificing utility. Future work should validate our findings by extending this analysis to Double Randomized Response to determine whether synthetic feature generation actually prevents the proposed subset selection attacks.

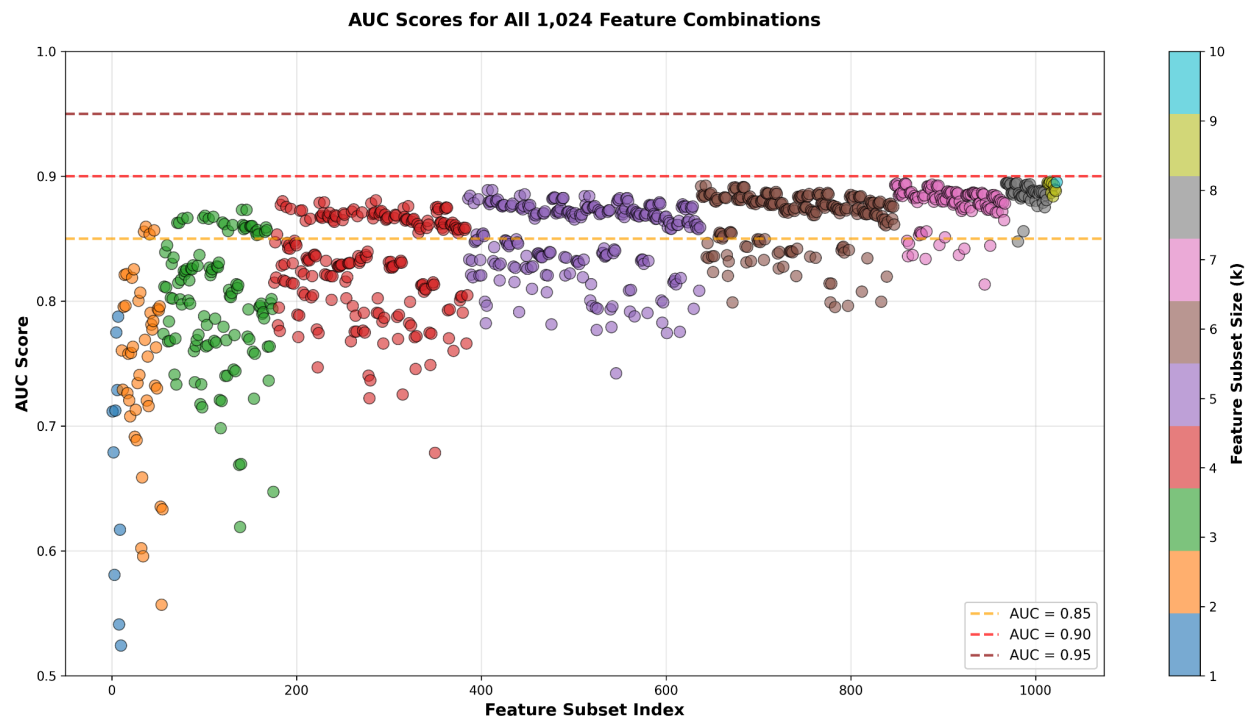
Ultimately, this research has highlighted a key insight that privacy protection for dataset releases requires auditing feature-label correlation strength alongside the sensitivity of its labels. By quantifying which specific feature combinations drive label inference success, this work provides a basis of evidence for what privacy mechanisms must be taken into account for publicized datasets, as a means of ensuring label confidentiality.

References

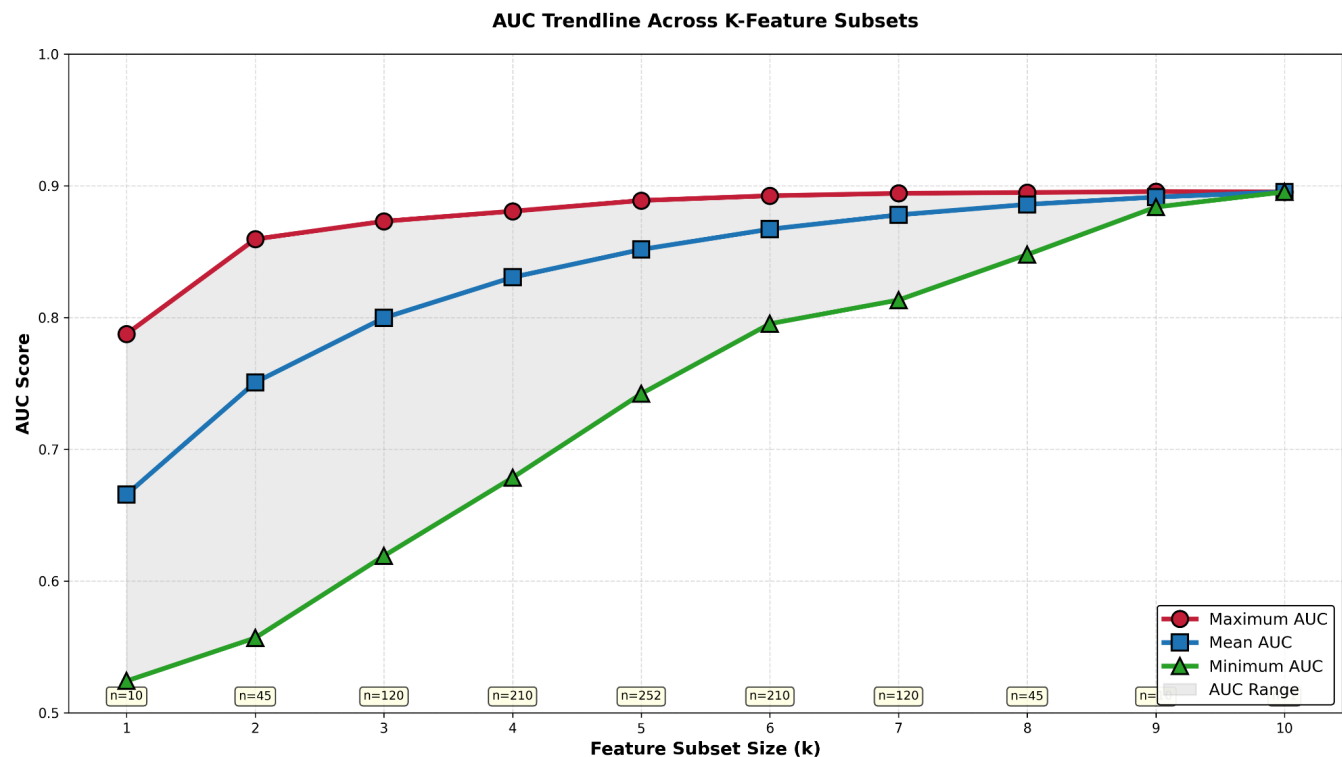
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A. Appendix

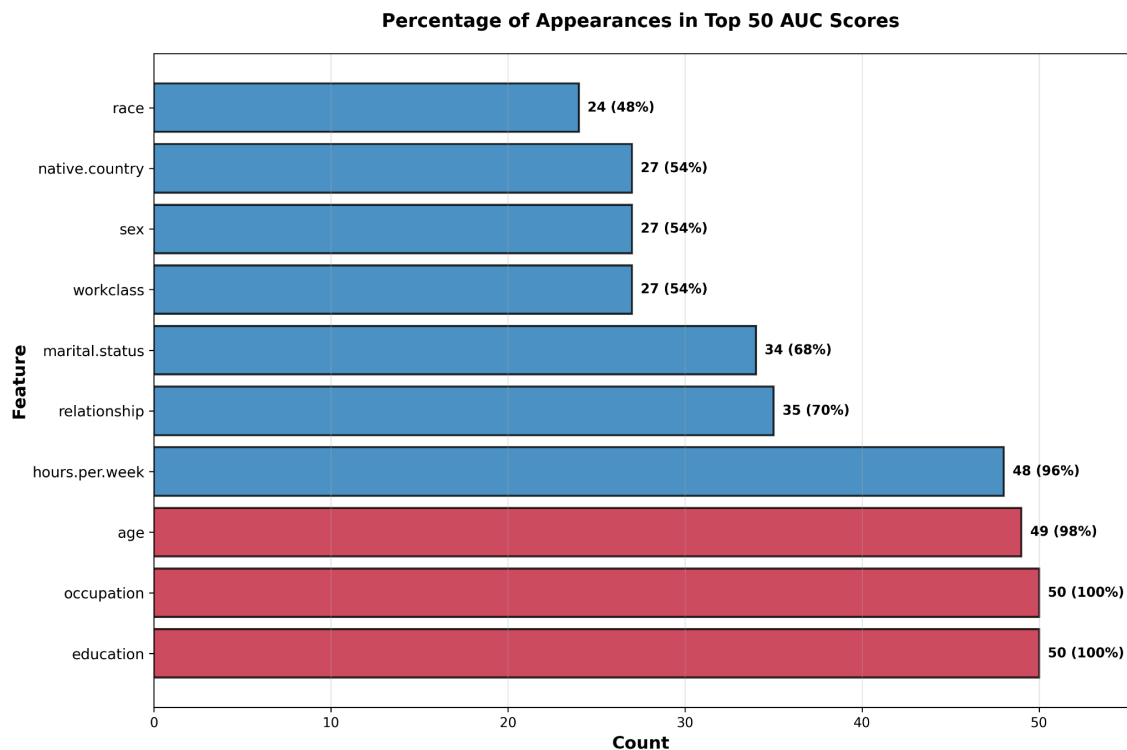
A.1 - Distribution of AUC Scores for All Feature Combinations



A.2 - AUC Score Trendline



A.3 - Most Frequent Features in Highest AUC Scores



A.4 - Source Code and Auxiliary Files

Jupyter notebooks used for model training/graph creation, full resolution images of the graphs, and csv files of Adult Income Dataset and model training outputs can be accessed via: <https://github.com/Martin8156/Label-Inference-on-Feature-Subsets>